



EduRev

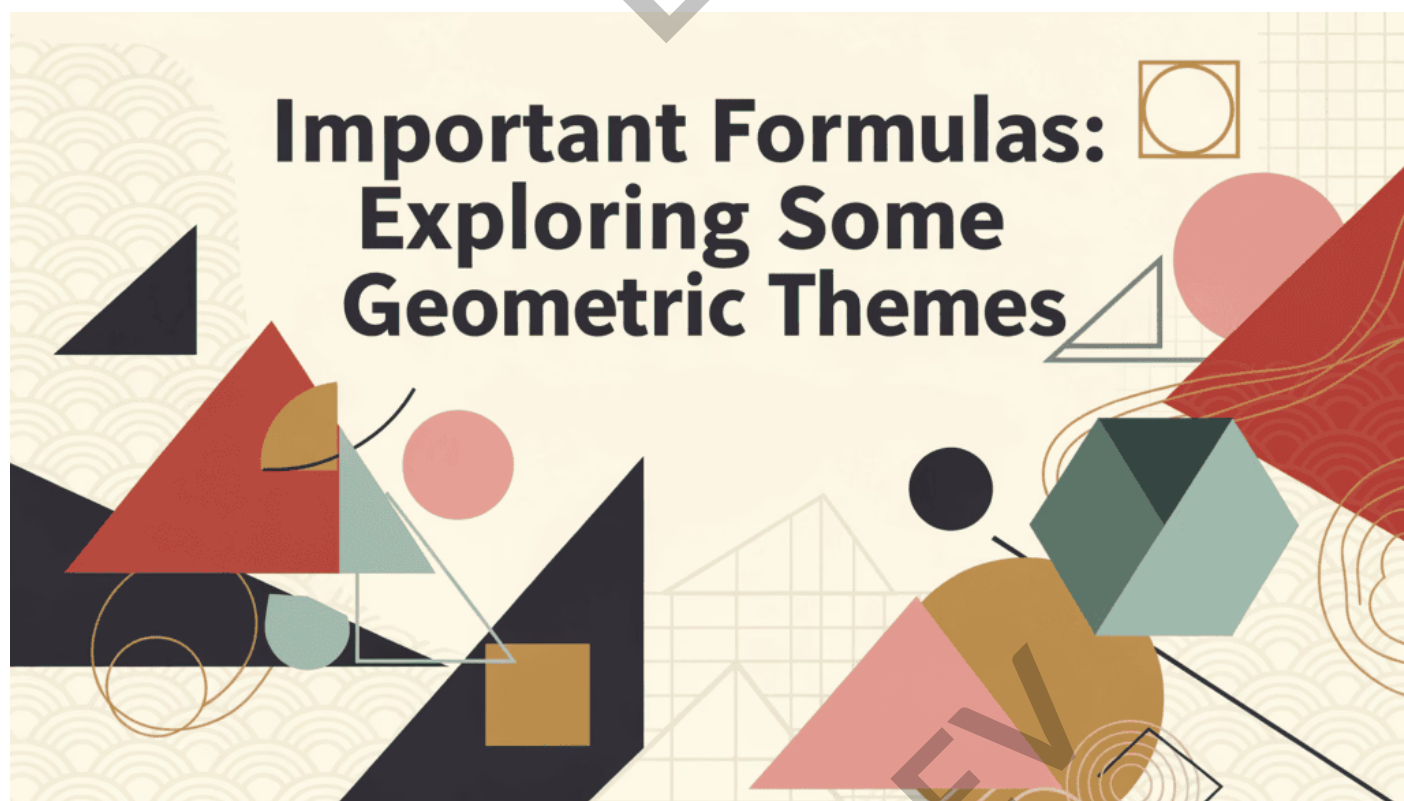
Important Formulas: Exploring Some Geometric Themes

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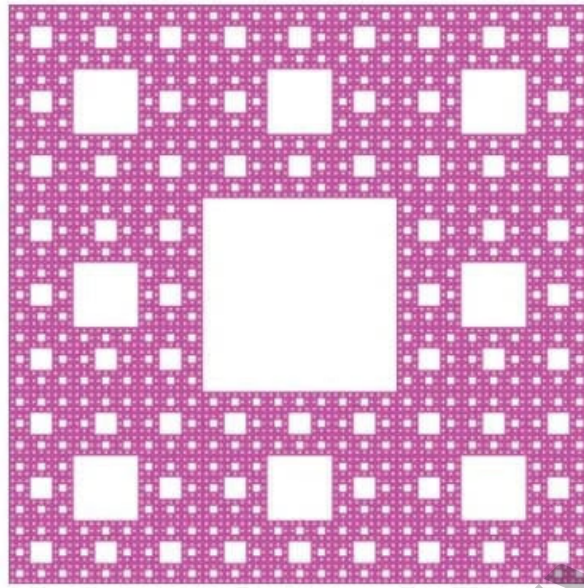
Important Formulas: Exploring Some Geometric Themes



Fractals

Sierpinski Carpet

A fractal created by repeatedly dividing a square into 9 smaller squares and removing the central square.



Pattern formulas

Let:

R_n be the number of remaining squares at step n .

H_n be the number of holes at step n .

Remaining squares

Recursive relation:

$$R_{n+1} = 8R_n.$$

Each remaining square is divided into 9 smaller squares. Eight remain and the central one becomes a hole.

General formula:

$$R_n = 8^n.$$

Initial values:

$$R_0 = 1$$

$$R_1 = 8$$

$$R_2 = 8^2 = 64$$

$$R_3 = 8^3 = 512$$

Number of holes

Recursive relation:

$$H_{n+1} = H_n + R_n.$$

Each remaining square creates one new hole at the next step.

All the holes created in earlier steps remain.

Pattern:

$$H_0 = 0$$

$$H_1 = 1$$

$$H_2 = 1 + 8 = 9$$

$$H_3 = 1 + 8 + 8^2 = 73$$

$$H_4 = 1 + 8 + 8^2 + 8^3 = 585$$

General formula:

$$H_n = 1 + 8 + 8^2 + \dots + 8^{n-1}$$

$$H_n = \frac{8^n - 1}{7}, \text{ using the sum of a geometric series.}$$

Sierpinski Triangle or Gasket

A fractal created by dividing an equilateral triangle into 4 identical triangles and removing the central triangle.



Pattern formulas

Let:

R_n be the number of remaining triangles at step n .

H_n be the number of holes at step n .

Remaining triangles**Recursive relation:**

$$R_{n+1} = 3R_n.$$

Each remaining triangle produces 3 smaller remaining triangles.

General formula:

$$R_n = 3^n.$$

Initial values:

$$R_0 = 1$$

$$R_1 = 3$$

$$R_2 = 3^2 = 9$$

$$R_3 = 3^3 = 27$$

Number of holes**Recursive relation:**

$$H_{n+1} = H_n + R_n.$$

The new hole is created from each remaining triangle, while all previous holes remain.

Pattern:

$$H_0 = 0$$

$$H_1 = 1$$

$$H_2 = 1 + 3 = 4$$

$$H_3 = 1 + 3 + 3^2 = 13$$

General formula:

$$H_n = \frac{3^n - 1}{2}.$$

Area remaining

If the initial area is 1 square unit:

At step n :

The area removed at each step is $\frac{1}{4}$ of the area remaining at the previous step.

The remaining area is $\left(\frac{3}{4}\right)^n$ square units.

Explanation: At each step, 1 triangle out of 4 is removed, so $\frac{3}{4}$ of the area remains.

Koch Snowflake

A fractal starting with an equilateral triangle, where each side is divided into 3 parts, and an equilateral triangle is raised over the middle part.



Number of sides

Recursive relation:

Each side is replaced by 4 smaller sides, each having one-third of the original length.

Formula:

The number of sides at step n is 3×4^n .

Initial values:

Step 0: 3 sides, representing the original triangle.

Step 1: $3 \times 4 = 12$ sides.

Step 2: $3 \times 4^2 = 48$ sides.

Step 3: $3 \times 4^3 = 192$ sides.

Perimeter

If the initial triangle has side length 1 unit:

At each step:

Each side of length L becomes 4 sides, each of length $\frac{L}{3}$.

Formula:

The perimeter at step n is $3\left(\frac{4}{3}\right)^n$ units.

Initial values:

Step 0: $P_0 = 3$ units.

Step 1: $P_1 = 3\left(\frac{4}{3}\right) = 4$ units.

Step 2: $P_2 = 3\left(\frac{4}{3}\right)^2 = \frac{16}{3} \approx 5.33$ units.

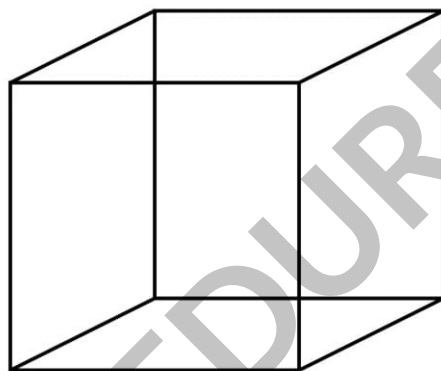
Step 3: $P_3 = 3\left(\frac{4}{3}\right)^3 = \frac{64}{9} \approx 7.11$ units.

Key property: The perimeter increases without bound as $n \rightarrow \infty$.

Solids: Faces, Edges and Vertices

A **face** is a flat surface of a solid. An **edge** is a line segment where two faces meet. A **vertex** is a point where edges meet.

Cube or Cuboid

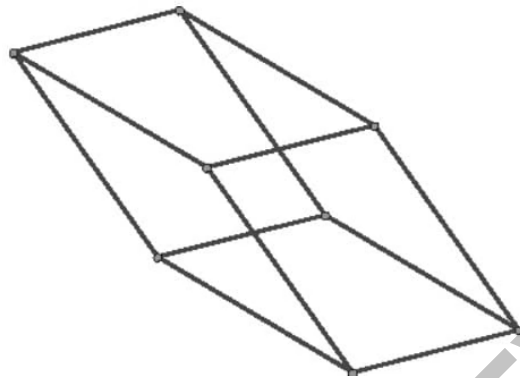


Faces: 6

Edges: 12

Vertices: 8

Prisms with an n -sided polygonal base



Triangular prism, where $n = 3$:

Faces: 5

Edges: 9

Vertices: 6

General n -sided prism:

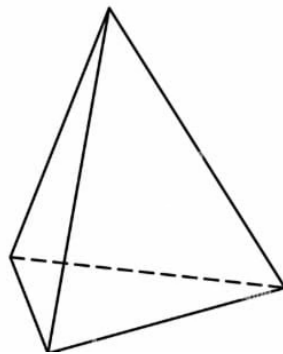
Faces: $n + 2$

Edges: $3n$

Vertices: $2n$

Pyramids with an n -sided polygonal base

Triangular pyramid or tetrahedron, where $n = 3$:

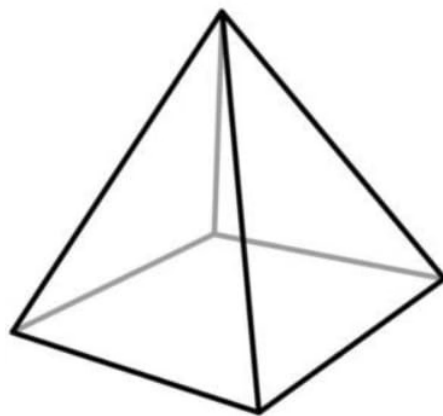


Faces: 4

Edges: 6

Vertices: 4

Square pyramid, where $n = 4$:



Faces: 5

Edges: 8

Vertices: 5

General n -sided pyramid:

Faces: $n + 1$

Edges: $2n$

Vertices: $n + 1$

Euler's relation for polyhedra

For a closed, convex polyhedron, the numbers of faces, vertices and edges are related by:

$$F + V - E = 2$$

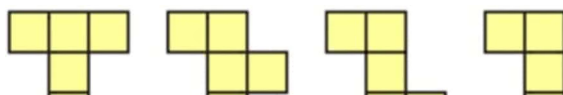
Here, F is the number of faces, V is the number of vertices and E is the number of edges.

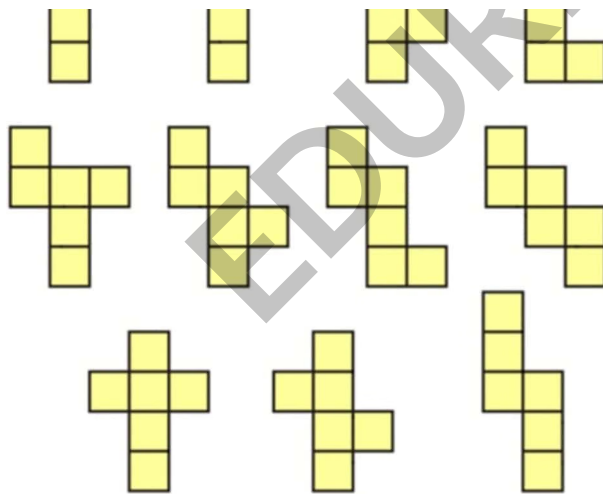
For a cube, $6 + 8 - 12 = 2$.

Nets of Solids

A **net** is a two-dimensional arrangement of the faces of a three-dimensional solid. When the arrangement is folded along its edges, it forms the solid.

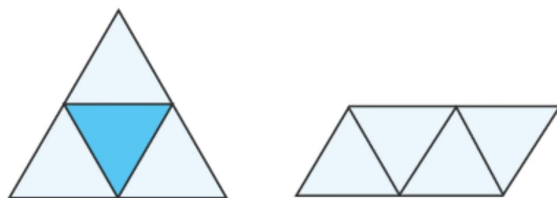
Cube





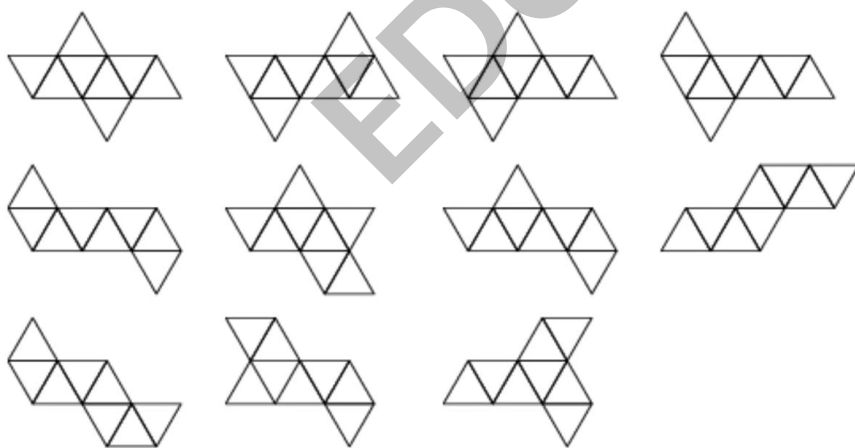
Total possible nets: 11, counting different structures but not rotations or reflections.

Regular tetrahedron



Total possible nets: 2.

Octahedron



Total possible nets: 11.

Dodecahedron

A dodecahedron has 12 pentagonal faces.

Total possible nets: 43,380.

Shortest Path on a Cuboid Surface

Method: Using nets

The shortest path on the surface of a cuboid can be found by:

1. **Unfolding the cuboid** into a net.
2. **Drawing a straight line** between the two points on the net.
3. **Calculating the distance** using the Baudhāyana-Pythagoras Theorem.

Formula

For the right triangle formed on the net:

$$d^2 = a^2 + b^2$$

Here, d is the shortest distance, while a and b are the perpendicular distances on the unfolded net.

Example

A cuboid has dimensions $6 \text{ cm} \times 12 \text{ cm} \times 30 \text{ cm}$. An ant is at one corner and a laddu is at the opposite corner on the back face.

After unfolding the cuboid:

One dimension is 24 cm.

The other dimension is 32 cm.

$$d^2 = 24^2 + 32^2 = 576 + 1024 = 1600$$

$$d = \sqrt{1600} = 40 \text{ cm}$$

Important note

The way a cuboid is unfolded matters. Different unfoldings may give different path lengths, so all suitable unfoldings should be checked to find the true shortest path.

Projections

The projection of a point P on a plane M is the point O where a line perpendicular from P meets the plane M .

Key properties

Projection of a line:

If a line has length l and its projection has length p , then $p \leq l$.

The projection length is equal to the actual length, $p = l$, when the line segment is parallel to the projection plane.

Projection of parallel lines:

Parallel lines generally project to parallel lines.

In a special position, their projections may coincide or become degenerate. Apart from this case, parallelism is preserved under projection.

Projection of polygons:

A parallelogram projects to a parallelogram when its projection is non-degenerate.

The projection of an n -sided polygon depends on its orientation with respect to the projection plane.

Three standard views

Front view: The projection on a vertical plane.

Top view: The projection on a horizontal plane.

Side view: The projection on a side or vertical plane.

Together, these three views provide important information about the shape and measurements of a three-dimensional object.

Isometric Projections

An **isometric projection** is an orientation in which the three principal directions of a solid have equal scale, so the projected edges of a cube have equal length.

Key property

Isometric means "equal measure".

All edge projections along the three principal directions have the same length for

equal actual lengths.

In an ideal isometric projection, a cube is commonly seen with a hexagonal outer outline, while its visible faces appear as parallelograms.

Three principal axes

On an isometric grid, the three principal axes are:

Height axis: The vertical direction.

Depth axis: One diagonal direction.

Length axis: The other diagonal direction.

Drawing guidelines

1. Equal unit distances along all three axes are represented by equal projected lengths.
2. Parallel edges in three-dimensional space remain parallel in the drawing.
3. The three families of parallel edges correspond to the three orientations of the isometric grid.

Geometric Series Connection

Many fractal formulas use a **geometric series**. In a geometric series, each term is obtained by multiplying the preceding term by a fixed number called the common ratio.

Sum formula:

$$\text{For } r \neq 1, 1 + r + r^2 + \dots + r^{n-1} = \frac{r^n - 1}{r - 1}.$$

Examples:

Sierpinski Carpet holes: $r = 8$.

$$H_n = 1 + 8 + 8^2 + \dots + 8^{n-1} = \frac{8^n - 1}{7}$$

Sierpinski Triangle holes: $r = 3$.

$$H_n = 1 + 3 + 3^2 + \dots + 3^{n-1} = \frac{3^n - 1}{2}$$

